

The Chebyshev–Lyapunov Certificate

Robust Stability, Preventive Diagnostics, Latent-State Identification, and Minimum-Cost Control for the Unified Three-Republic System

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Abstract

The first paper in this programme established a unified three-Republic equilibrium incorporating algebraic consistency, strategic equilibrium, dynamic stability, welfare-strategic optimality, latent-state identification, stochastic risk, and robust viability. The present paper develops the corresponding computational and certification layer. Its principal contribution is a Chebyshev–Lyapunov framework for converting continuous parameter uncertainty into finite, reproducible, and independently verifiable stability certificates. The strategically reduced Jacobian is represented by Chebyshev expansions over a normalized uncertainty domain, while parameter-dependent Lyapunov matrices are represented in the same basis. Positive-definiteness and Lyapunov decrease are then certified using coefficient bounds together with explicit approximation-error margins. The framework distinguishes numerical verification from mathematical certification and explicitly prevents a failed sufficient certificate from being interpreted as proof of instability. Common and asymmetric Republic modes receive separate certificates, permitting early detection of systemic and divergence-driven instability while the system remains dynamically stable. The resulting diagnostic engine combines strategic-feedback margins, Lyapunov margins, algebraic separation, latent-state observability, welfare curvature, transient amplification, and finite-horizon regime-transition probabilities. A minimum-cost preventive control problem is then formulated to restore the system to a robustly certified safe region before an instability boundary is crossed. Connections to robust control, economics, finance, physics, chemistry, biology, psychology, psychiatry, political science, diplomacy, international relations, warfare economics, statistics, computer science, data science, and artificial intelligence are developed as applications and interpretations of the common mathematical structure. The paper concludes with a reproducible algorithm and a unit-test architecture suitable for implementation in scientific software and continuous integration.

The paper ends with “The End”

Keywords: Chebyshev polynomials; Lyapunov stability; robust control; three-Republic system; dynamic games; systemic risk; preventive diagnostics; latent states; observability; welfare economics; stochastic stability; polynomial approximation; machine-checkable certificates.

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1 Introduction

The preceding Unified Three-Republic Theorem established a coupled equilibrium architecture in which algebraic feasibility, strategic interaction, dynamic stability, welfare, latent-state identification, uncertainty, and preventive control are treated as components of one system.

The next problem is computational.

A theorem may establish that a certificate exists without specifying how a researcher or policy-maker should calculate that certificate over a continuous uncertainty domain. Direct grid evaluation is insufficient because a finite grid cannot, without additional assumptions, establish a continuum inequality. Pointwise eigenvalue calculations are similarly insufficient for a robust theorem when the Jacobian depends on uncertain parameters.

The present paper addresses this gap.

The principal construction is a Chebyshev–Lyapunov certificate. Chebyshev polynomials provide a numerically effective basis on compact intervals, while Lyapunov functions provide a constructive stability certificate. The two are combined so that a parameter-dependent Lyapunov matrix can be represented by finitely many coefficients.

The basic hierarchy is

$$\text{model} \longrightarrow \text{approximation} \longrightarrow \text{error bound} \longrightarrow \text{certificate}. \quad (1)$$

The distinction is fundamental. A numerical approximation is evidence. An error-bounded approximation can become a certificate.

2 The Unified Three-Republic System

Let the state be

$$s_t = (r_{1t} \ r_{2t} \ r_{3t} \ p_t), \quad u_t = (u_{1t} \ u_{2t} \ u_{3t}). \quad (2)$$

Let z_t denote latent structural states. The stochastic system is

$$s_{t+1} = T(s_t, u_t, \theta, \varepsilon_{t+1}), \quad (3)$$

$$u_t^* = R(u_t^*, s_t, a_t, \theta, \xi_{t+1}), \quad (4)$$

$$z_{t+1} = F_z(z_t, s_t, u_t, \theta, \eta_{t+1}), \quad (5)$$

and

$$Y_t = H(s_t, z_t, \theta) + \nu_t. \quad (6)$$

The structural equilibrium restrictions are

$$\Phi(s, u; \theta) = 0. \quad (7)$$

The Republic-specific states possess the symmetric invariants

$$r_A = r_1 + r_2 + r_3, \quad (8)$$

$$r_B = r_1 r_2 + r_1 r_3 + r_2 r_3, \quad (9)$$

and

$$r_C = r_1 r_2 r_3. \quad (10)$$

They are roots of

$$P(\lambda) = \lambda^3 - r_A \lambda^2 + r_B \lambda - r_C. \quad (11)$$

The corresponding discriminant is

$$\Delta = r_A^2 r_B^2 - 4r_B^3 - 4r_A^3 r_C - 27r_C^2 + 18r_A r_B r_C. \quad (12)$$

A normalized algebraic separation margin is

$$m_\Delta = \frac{\Delta}{(1 + |r_A| + |r_B| + |r_C|)^6}. \quad (13)$$

3 Strategic Reduction

Let the primitive transition map be

$$\tilde{T}(s, u; \theta). \quad (14)$$

The strategic equilibrium satisfies

$$u^* = R(u^*, s; \theta). \quad (15)$$

Define

$$K(\theta) = D_u R(u^*, s; \theta). \quad (16)$$

If

$$\rho(K) < 1, \quad (17)$$

then

$$(I - K)^{-1} = I + K + K^2 + K^3 + \dots. \quad (18)$$

Differentiating the strategic fixed point gives

$$D_s u^* = (I - D_u R)^{-1} D_s R. \quad (19)$$

Consequently,

$$\boxed{D_s T^* = D_s \tilde{T} + D_u \tilde{T} (I - D_u R)^{-1} D_s R.} \quad (20)$$

This identity is the fundamental bridge between strategic interaction and dynamic stability.

Theorem 3.1 (Strategic amplification). *Suppose $I - D_u R$ is nonsingular. Then the sensitivity of the strategically reduced transition map contains the feedback multiplier $(I - D_u R)^{-1}$. Consequently, an apparently moderate direct state response can become dynamically large when strategic feedback approaches singularity.*

Proof. The strategic equilibrium satisfies $u^* - R(u^*, s) = 0$. Differentiation with respect to s gives

$$D_s u^* - D_u R D_s u^* - D_s R = 0. \quad (21)$$

Rearrangement gives

$$(I - D_u R) D_s u^* = D_s R. \quad (22)$$

Premultiplication by $(I - D_u R)^{-1}$ gives the result. Substitution into the chain rule for T^* completes the proof. $\square$

4 Dynamic Stability

The equilibrium satisfies

$$s^* = T^*(s^*). \quad (23)$$

The local reduced Jacobian is

$$J^* = D_s T^*(s^*). \quad (24)$$

Local discrete-time asymptotic stability follows from

$$\rho(J^*) < 1. \quad (25)$$

However, spectral stability is not equivalent to absence of transient amplification. Define

$$\mathcal{A}_H = \max_{1 \leq h \leq H} |(J^*)^h|. \quad (26)$$

A system may satisfy

$$\rho(J^*) < 1 \quad (27)$$

while having

$$\mathcal{A}_H \gg 1. \quad (28)$$

Thus asymptotic stability and short-run resilience must be reported separately.

5 Common and Asymmetric Modes

Define

$$\mathbf{1} = \begin{pmatrix} 1 & 1 & 1 \end{pmatrix}, \quad \Pi_C = \frac{1}{3} \mathbf{1} \mathbf{1}^\top, \quad (29)$$

and

$$\Pi_A = I - \Pi_C. \quad (30)$$

Then

$$\Pi_C^2 = \Pi_C, \quad \Pi_A^2 = \Pi_A, \quad \Pi_C \Pi_A = 0. \quad (31)$$

The common and asymmetric Jacobians are

$$J_C = \Pi_C J \Pi_C, \quad (32)$$

and

$$J_A = \Pi_A J \Pi_A. \quad (33)$$

Define the spectral margins

$$m_C = 1 - \rho(J_C), \quad (34)$$

and

$$m_A = 1 - \rho(J_A). \quad (35)$$

The four principal regimes are

Regime	Common mode	Asymmetric mode
I	Stable	Stable
II	Unstable	Stable
III	Stable	Unstable
IV	Unstable	Unstable

The principal advantage of this decomposition is that systemic deterioration and Republic divergence are separately observable.

6 Welfare-Strategic Optimality

Let

$$W^\omega = \sum_{i=1}^3 \omega_i V_i, \quad \omega_i > 0. \quad (36)$$

An interior welfare optimum satisfies

$$\nabla_u W^\omega = 0. \quad (37)$$

Strict local optimality requires

$$D_{uu}^2 W^\omega \prec 0 \quad (38)$$

on the tangent space of active constraints.

Define the welfare curvature margin

$$m_W = \lambda_{\min} \left(-D^2 W^\omega \big|_* T * \mathcal{C} \right). \quad (39)$$

Then

$$m_W > 0 \quad (40)$$

is a sufficient local strict-curvature condition.

7 Latent-State Identification

The observation process is

$$Y_t = H(q_t, z_t, \theta) + \nu_t, \quad (41)$$

where

$$q_t = (r_{A,t} \ r_{B,t} \ r_{C,t} \ p_t). \quad (42)$$

The local observability matrix is

$$\mathcal{O}_z = \begin{pmatrix} H_z & H_z F_z & H_z F_z^2 & \vdots & H_z F_z^{k-1} \end{pmatrix}. \quad (43)$$

Local identification requires

$$\text{rank } \mathcal{O}_z = k. \quad (44)$$

The numerical identification margin is

$$m_I = \sigma_{\min}(\mathcal{O}_z). \quad (45)$$

As

$$m_I \rightarrow 0, \quad (46)$$

latent-state recovery becomes ill-conditioned.

8 Chebyshev Polynomials

The first-kind Chebyshev polynomials are defined recursively by

$$T_0(x) = 1, \quad T_1(x) = x, \quad (47)$$

and

$$T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x). \quad (48)$$

Their trigonometric representation is

$$T_n(\cos \theta) = \cos(n\theta). \quad (49)$$

For

$$x \in [-1, 1], \quad (50)$$

we have

$$|T_n(x)| \leq 1. \quad (51)$$

This last inequality is central to the finite certificate.

8.1 Parameter normalization

For a scalar uncertain parameter,

$$x = \frac{2\theta - \theta^+ - \theta^-}{\theta^+ - \theta^-}. \quad (52)$$

Thus

$$\theta \in [\theta^-, \theta^+] \iff x \in [-1, 1]. \quad (53)$$

For d parameters,

$$x_i = \frac{2\theta_i - \theta_i^+ - \theta_i^-}{\theta_i^+ - \theta_i^-}, \quad i = 1, \dots, d. \quad (54)$$

Define

$$T_{\mathbf{k}}(x) = \prod_{i=1}^d T_{k_i}(x_i). \quad (55)$$

9 Finite-Degree Chebyshev Representation

The reduced Jacobian is approximated by

$$\hat{J}_N(x) = \sum_{\mathbf{k} \in \mathcal{K}_N} J_{\mathbf{k}} T_{\mathbf{k}}(x). \quad (56)$$

A parameter-dependent Lyapunov matrix is represented as

$$\hat{P}_N(x) = \sum_{\mathbf{k} \in \mathcal{K}_N} P_{\mathbf{k}} T_{\mathbf{k}}(x). \quad (57)$$

A total-degree index set is

$$\mathcal{K}_N^{\text{TD}} = \left\{ \mathbf{k} : \sum_{i=1}^d k_i \leq N \right\}. \quad (58)$$

Its cardinality is

$$|\mathcal{K}_N^{\text{TD}}| = \binom{N+d}{d}. \quad (59)$$

This avoids the full tensor-product explosion.

10 Chebyshev Sampling

The canonical nodes are

$$x_j = \cos\left(\frac{(2j-1)\pi}{2M}\right), \quad j = 1, \dots, M. \quad (60)$$

For an interval $[\theta^-, \theta^+]$,

$$\theta_j = \frac{\theta^+ + \theta^-}{2} + \frac{\theta^+ - \theta^-}{2} x_j. \quad (61)$$

The discrete coefficient approximation is

$$c_k \approx \frac{2}{M} \sum_{j=1}^M f(x_j) \cos\left(k \frac{(2j-1)\pi}{2M}\right). \quad (62)$$

Matrix coefficients are obtained componentwise.

11 Clenshaw Evaluation

Direct evaluation of a high-degree Chebyshev series is unnecessary.

Set

$$b_{N+1} = b_{N+2} = 0, \quad (63)$$

and recursively define

$$b_k = 2xb_{k+1} - b_{k+2} + c_k, \quad k = N, \dots, 0. \quad (64)$$

The polynomial is then evaluated by

$$p(x) = b_0 - xb_2. \quad (65)$$

This permits stable repeated evaluation of diagnostic surfaces.

12 Parameter-Dependent Lyapunov Stability

Consider the perturbation dynamics

$$\delta s_{t+1} = J(x_t)\delta s_t. \quad (66)$$

Let

$$V(x, \delta s) = \delta s^\top P(x) \delta s, \quad P(x) \succ 0. \quad (67)$$

The discrete Lyapunov decrease matrix is

$$Q(x) = P(x) - J(x)^\top P(x) J(x). \quad (68)$$

Uniform robust stability is certified by

$$P(x) \succ 0 \quad (69)$$

and

$$Q(x) \succ 0 \quad (70)$$

for every admissible x .

Theorem 12.1 (Uniform Lyapunov stability). *Suppose there exist constants $0 < \underline{p} \leq \bar{p}$ and $\underline{q} > 0$ such that*

$$\underline{p}I \preceq P(x) \preceq \bar{p}I \quad (71)$$

and

$$Q(x) \succeq \underline{q}I \quad (72)$$

for every $x \in [-1, 1]^d$. Then the uncertain linear system

$$\delta s_{t+1} = J(x_t)\delta s_t \quad (73)$$

is uniformly exponentially stable.

Proof. The Lyapunov function satisfies

$$\underline{p}|\delta s_t|^2 \leq V_t \leq \bar{p}|\delta s_t|^2. \quad (74)$$

Moreover,

$$V_{t+1} - V_t = \delta s_t^\top (J(x_t)^\top P(x_t) J(x_t) - P(x_t)) \delta s_t = -\delta s_t^\top Q(x_t) \delta s_t \leq -\underline{q}|\delta s_t|^2. \quad (75)$$

Using the upper bound on V_t gives

$$V_{t+1} \leq \left(1 - \frac{q}{p}\right) V_t. \quad (76)$$

Hence V_t and therefore $\|\delta s_t\|$ decay geometrically. $\square$

13 Finite Positivity Certificates

Suppose

$$\hat{P}_N(x) = P_0 + \sum_{\mathbf{k} \neq 0} P_{\mathbf{k}} T_{\mathbf{k}}(x). \quad (77)$$

Because

$$|T_{\mathbf{k}}(x)| \leq 1, \quad (78)$$

we obtain

$$\left\| \sum_{\mathbf{k} \neq 0} P_{\mathbf{k}} T_{\mathbf{k}}(x) \right\|_2 \leq \sum_{\mathbf{k} \neq 0} \|P_{\mathbf{k}}\|_2. \quad (79)$$

Therefore define

$$\gamma_P = \lambda_{\min}(P_0) - \sum_{\mathbf{k} \neq 0} \|P_{\mathbf{k}}\|_2. \quad (80)$$

Proposition 13.1 (Chebyshev positivity certificate). *If*

$$\gamma_P > 0, \quad (81)$$

then

$$\hat{P}_N(x) \succ 0 \quad (82)$$

for every $x \in [-1, 1]^d$.

Proof. For every unit vector v ,

$$v^\top \hat{P}_N(x) v \geq \lambda_{\min}(P_0) - \left\| \sum_{\mathbf{k} \neq 0} P_{\mathbf{k}} T_{\mathbf{k}}(x) \right\|_2 \quad (83)$$

$$\geq \lambda_{\min}(P_0) - \sum_{\mathbf{k} \neq 0} \|P_{\mathbf{k}}\|_2 \quad (84)$$

$$= \gamma_P > 0. \quad (85)$$

Hence $\hat{P}_N(x)$ is positive definite everywhere on the domain. $\square$

The same construction applies to

$$\hat{Q}_N(x) = \hat{P}_N(x) - \hat{J}_N(x)^\top \hat{P}_N(x) \hat{J}_N(x). \quad (86)$$

Define

$$\gamma_Q = \lambda_{\min}(Q_0) - \sum_{\mathbf{k} \neq 0} \|Q_{\mathbf{k}}\|_2. \quad (87)$$

Then

$$\gamma_Q > 0 \quad (88)$$

is a sufficient uniform certificate for

$$\hat{Q}_N(x) \succ 0. \quad (89)$$

14 Approximation Error

A finite Chebyshev representation is not exact in general.

Suppose

$$\|P - \hat{P}_N\|_\infty \leq \epsilon_P \quad (90)$$

and

$$\|Q - \hat{Q}_N\|_\infty \leq \epsilon_Q. \quad (91)$$

Then Weyl's eigenvalue perturbation inequality implies

$$\lambda_{\min}(P(x)) \geq \lambda_{\min}(\hat{P}_N(x)) - \epsilon_P. \quad (92)$$

Likewise,

$$\lambda_{\min}(Q(x)) \geq \lambda_{\min}(\hat{Q}_N(x)) - \epsilon_Q. \quad (93)$$

Therefore the error-aware certificate is

$$\boxed{\inf_x \lambda_{\min}(\hat{P}_N(x)) > \epsilon_P} \quad (94)$$

and

$$\boxed{\inf_x \lambda_{\min}(\hat{Q}_N(x)) > \epsilon_Q.} \quad (95)$$

Theorem 14.1 (Error-corrected Chebyshev–Lyapunov certificate). *Suppose the approximation errors satisfy the preceding bounds. If*

$$\inf_x \lambda_{\min}(\hat{P}_N(x)) > \epsilon_P \quad (96)$$

and

$$\inf_x \lambda_{\min}(\hat{Q}_N(x)) > \epsilon_Q, \quad (97)$$

then the exact matrices $P(x)$ and $Q(x)$ are positive definite throughout the uncertainty domain. Consequently, the underlying uncertain linear system is uniformly Lyapunov stable.

Proof. The eigenvalue perturbation inequalities imply

$$\lambda_{\min}(P(x)) > 0 \quad (98)$$

and

$$\lambda_{\min}(Q(x)) > 0 \quad (99)$$

uniformly. The preceding Lyapunov theorem then applies. $\square$

15 Common and Asymmetric Lyapunov Certificates

Define parameter-dependent matrices

$$P_C(x) = \sum_{\mathbf{k}} P_{C,\mathbf{k}} T_{\mathbf{k}}(x), \quad (100)$$

and

$$P_A(x) = \sum_{\mathbf{k}} P_{A,\mathbf{k}} T_{\mathbf{k}}(x). \quad (101)$$

The corresponding Lyapunov decrease matrices are

$$Q_C(x) = P_C(x) - J_C(x)^\top P_C(x) J_C(x), \quad (102)$$

and

$$Q_A(x) = P_A(x) - J_A(x)^\top P_A(x) J_A(x). \quad (103)$$

Define

$$\gamma_C = \inf_x \lambda_{\min}(Q_C(x)), \quad (104)$$

and

$$\gamma_A = \inf_x \lambda_{\min}(Q_A(x)). \quad (105)$$

The certified Regime-I condition is

$$\gamma_C > 0, \quad \gamma_A > 0. \quad (106)$$

This produces independent common-mode and asymmetric-mode certificates.

16 Preventive Stability Margins

The spectral margins are

$$m_S = 1 - \rho(D_u R), \quad (107)$$

$$m_C = 1 - \rho(J_C), \quad (108)$$

and

$$m_A = 1 - \rho(J_A). \quad (109)$$

The complete diagnostic vector is

$$\mathcal{D}_t = \begin{pmatrix} m_S(t) \\ m_C(t) \\ m_A(t) \\ \gamma_C(t) \\ \gamma_A(t) \\ m_I(t) \\ m_\Delta(t) \\ m_W(t) \\ \mathcal{A}_H(t) \end{pmatrix}. \quad (110)$$

A system can therefore be stable while entering a warning state if one or more margins approaches its policy threshold.

17 Warning States Inside Regime I

Choose positive thresholds

$$\tau_S, \tau_C, \tau_A, \tau_I, \tau_\Delta, \tau_W, \tau_{\gamma_C}, \tau_{\gamma_A}. \quad (111)$$

The robust core of Regime I satisfies

$$m_j > \tau_j \quad (112)$$

for every required margin.

The warning region is

$$\mathcal{W} = \{s : 0 < m_j \leq \tau_j \text{ for at least one } j\}. \quad (113)$$

The endangered but still stable region is therefore

$$\mathcal{E} = \{s : 0 < m_C, m_A \text{ and } s \in \mathcal{W}\}. \quad (114)$$

The system can consequently be classified as

State	Mathematical condition	Operational interpretation
I_0	All margins comfortably positive	Robust Regime I
I_1	Strategic margin deteriorating	Strategic warning
I_2	γ_C or m_C deteriorating	Common-mode warning
I_3	γ_A or m_A deteriorating	Asymmetric warning
I_4	Multiple margins deteriorating	Compound warning
II	Common-mode instability	Systemic instability
III	Asymmetric instability	Divergence instability
IV	Both unstable	Compound instability

18 Diagnostic Forecasting

For any margin m_j , define

$$\dot{m}_j(t) \approx \frac{m_j(t) - m_j(t - \Delta t)}{\Delta t}. \quad (115)$$

If

$$\dot{m}_j(t) < 0, \quad (116)$$

a linearized boundary-crossing estimate is

$$\hat{\tau}_j = \frac{m_j(t)}{-\dot{m}_j(t)}. \quad (117)$$

The earliest estimated crossing is

$$\hat{\tau}_{\min} = \min_j \hat{\tau}_j. \quad (118)$$

This is a local forecast and should not be interpreted as a theorem without additional assumptions. A more robust forecast models the future margin itself by

$$m_j(t+h) \approx \sum_{k=0}^N c_{jk}(t) T_k(\tau_h), \quad (119)$$

where τ_h maps the forecast horizon into $[-1, 1]$.

19 Stochastic Risk

Let

$$\tau_k = \inf\{h \geq 1 : s_{t+h} \in \mathcal{R}_k\}. \quad (120)$$

Define

$$P_{I \rightarrow II}^{(H)} = \Pr(\tau_{II} \leq H \mid s_t), \quad (121)$$

$$P_{I \rightarrow III}^{(H)} = \Pr(\tau_{III} \leq H \mid s_t), \quad (122)$$

and

$$P_{I \rightarrow IV}^{(H)} = \Pr(\tau_{IV} \leq H \mid s_t). \quad (123)$$

The preventive risk vector is

$$\mathcal{P}_I^{(H)} = \begin{pmatrix} P_{I \rightarrow II}^{(H)} \\ P_{I \rightarrow III}^{(H)} \\ P_{I \rightarrow IV}^{(H)} \end{pmatrix}. \quad (124)$$

A policymaker may therefore distinguish a low-probability but catastrophic transition to Regime IV from a more probable but less severe transition to Regime II.

20 Robust Parameter Uncertainty

Let

$$\theta \in \Theta \quad (125)$$

and let $\mathcal{P}$ denote an admissible class of shock distributions. Define the robust probability

$$\underline{\Pr}(A) = \inf_{\theta \in \Theta} \inf_{P \in \mathcal{P}} P_\theta(A). \quad (126)$$

The robust safety requirement is

$$\boxed{\underline{\Pr}[s_t \in \mathcal{S}_{\text{safe}} \forall t \geq 0] \geq 1 - \alpha.} \quad (127)$$

The robust spectral conditions are

$$\sup_{\theta \in \Theta} \rho(D_u R(\theta)) \leq 1 - \delta_S, \quad (128)$$

and

$$\sup_{\theta \in \Theta} \rho(J(\theta)) \leq 1 - \delta_J. \quad (129)$$

The robust identification condition is

$$\inf_{\theta \in \Theta} \sigma_{\min}(\mathcal{O}_z(\theta)) \geq \delta_I > 0. \quad (130)$$

21 Robust Viability

Define the robust stochastic viability kernel

$$\mathcal{K}_\alpha^R = \{s_0 : \exists \pi \text{ such that } \underline{\Pr}^\pi[s_t \in \mathcal{S}_{\text{safe}} \forall t] \geq 1 - \alpha\}. \quad (131)$$

The three principal cases are

Region	Property	Meaning
$\text{int } \mathcal{K}_\alpha^R$	Robustly viable	Safe and controllably safe
$\mathcal{S}_{\text{safe}} \setminus \mathcal{K}_\alpha^R$	Nonviable	Currently safe but not sustainably controllable
$\mathcal{S} \setminus \mathcal{S}_{\text{safe}}$	Unsafe	Preventive boundary breached

The distinction between stability and viability is fundamental. A stable state may nevertheless be outside the robust viability kernel if future disturbances or constraints make continued safety impossible.

22 Minimum-Cost Preventive Control

Let a denote intervention.

The preventive control problem is

$$\begin{aligned}
& \min_{a, P_C, P_A} C(a) \\
& \text{subject to } \Phi(s^+, u^+; \theta) = 0, \\
& \quad u^+ = R(u^+, s^+, a, \theta), \\
& \quad P_C(x) \succ 0, \\
& \quad P_A(x) \succ 0, \\
& \quad Q_C(x) \succ 0, \\
& \quad Q_A(x) \succ 0, \\
& \quad m_j(x, a) \geq \bar{m}_j, \\
& \quad \underline{\Pr}[s_t \in \mathcal{S}_{\text{safe}} \forall t] \geq 1 - \alpha.
\end{aligned} \tag{132}$$

If the cost function is quadratic,

$$C(a) = a^\top Q a, \quad Q \succ 0, \tag{133}$$

then local preventive interventions can be computed through convex quadratic programming whenever the remaining constraints are affine or convexified.

23 Local Intervention Formula

Suppose a margin has local approximation

$$m_j(s^+, a) \approx m_j(s) + b_j^\top a. \tag{134}$$

The constraint is

$$b_j^\top a \geq \bar{m}_j - m_j. \tag{135}$$

The minimum quadratic intervention is

$$a^\dagger = \frac{\bar{m}_j - m_j}{b_j^\top Q^{-1} b_j} Q^{-1} b_j. \tag{136}$$

This solution is meaningful only when

$$\bar{m}_j > m_j. \tag{137}$$

If the constraint is already satisfied, the unconstrained minimum is

$$a^\dagger = 0. \tag{138}$$

24 Sequential Convexification

Because the complete robust problem can be nonconvex, define the local Jacobian approximation

$$J^{(q+1)} \approx J^{(q)} + D_a J^{(q)} \Delta a. \tag{139}$$

Likewise,

$$P^{(q+1)} \approx P^{(q)} + D_a P^{(q)} \Delta a. \tag{140}$$

At each iteration solve the local convex subproblem, update the intervention, and recompute the full nonlinear certificate.

The algorithm terminates only when

$$\|\Delta a\| \leq \varepsilon_a \quad (141)$$

and every required certificate inequality is satisfied.

25 A Machine-Checkable Certificate

Define

$$\mathfrak{C} = (C_{\text{alg}}, C_{\text{strat}}, C_{\text{dyn}}, C_{\text{welfare}}, C_{\text{id}}, C_{\text{robust}}, C_{\text{risk}}). \quad (142)$$

A complete certificate requires

$$C_{\text{alg}} = C_{\text{strat}} = C_{\text{dyn}} = C_{\text{welfare}} = C_{\text{id}} = C_{\text{robust}} = C_{\text{risk}} = 1. \quad (143)$$

The computational implementation must distinguish the following:

Output	Meaning
Certified safe	All sufficient certificate conditions pass
Certificate failed	At least one sufficient test fails
Numerically unstable	Direct numerical evidence indicates instability
Undetermined	Evidence is insufficient for classification

In particular,

$$\boxed{\text{certificate failure} \not\Rightarrow \text{instability.}} \quad (144)$$

A sufficient certificate can fail even when the system is stable.

26 Independent Verification

A reliable implementation should contain three independent verification layers.

26.1 Analytical layer

Verify

$$\Phi = 0, \quad (145)$$

$$u = R(u, s), \quad (146)$$

and all dimension and domain conditions.

26.2 Numerical layer

Compute

$$\rho(D_u R), \quad \rho(J_C), \quad \rho(J_A), \quad (147)$$

and direct numerical Lyapunov margins.

26.3 Certificate layer

Verify

$$\gamma_P > 0, \quad \gamma_{Q_C} > 0, \quad \gamma_{Q_A} > 0, \quad (148)$$

after accounting for approximation and numerical errors.

The three layers should agree before a production certificate is issued.

27 Convergence of the Chebyshev Representation

Let $\hat{f}_N$ denote the degree- N approximation.

Define

$$E_N = \|\hat{f}_{2N} - \hat{f}_N\|_\infty. \quad (149)$$

Accept the degree N representation only when

$$E_N \leq \varepsilon_{\text{approx}}. \quad (150)$$

This gives a reproducible stopping rule.

If the coefficients decay rapidly,

$$|c_k| \rightarrow 0, \quad (151)$$

then increasing N should eventually stabilize the representation. Slowly decaying coefficients provide a diagnostic for nonsmoothness, singularities, or inadequate model regularity.

28 Application to Economics and Finance

The framework can represent a macro-financial system in which state variables include output, inflation, debt, liquidity, leverage, asset prices, and risk premia.

For example,

$$s_t = \begin{pmatrix} Y_t \\ \pi_t \\ B_t \\ L_t \end{pmatrix} \quad (152)$$

can represent output, inflation, sovereign debt, and liquidity.

A financial feedback loop may take the form

$$\text{leverage} \rightarrow \text{asset prices} \rightarrow \text{collateral} \rightarrow \text{borrowing capacity} \rightarrow \text{leverage}. \quad (153)$$

The resulting feedback matrix is mathematically analogous to strategic feedback among the Republics.

The preventive objective is therefore not necessarily to maximize a contemporaneous economic statistic. It may instead minimize intervention cost subject to preservation of a robust dynamic region.

29 Physics

The Lyapunov function

$$V = \delta s^\top P \delta s \quad (154)$$

can be interpreted as a generalized perturbation energy.

The condition

$$V_{t+1} - V_t < 0 \quad (155)$$

represents dissipation.

Non-normal transient amplification is analogous to physical systems in which perturbation energy can temporarily increase despite eventual asymptotic decay.

The common and asymmetric decomposition is likewise analogous to normal-mode analysis.

30 Chemistry

A chemical reaction network may be represented as

$$x_{t+1} = T(x_t, u_t). \quad (156)$$

Conservation laws become algebraic constraints, equilibrium concentrations become fixed points, reaction intermediates become latent states, and catalytic or removal actions become controls.

The same distinction arises between

$$\text{equilibrium} \quad (157)$$

and

$$\text{stability of equilibrium}. \quad (158)$$

A chemically feasible equilibrium may nevertheless be dynamically unstable.

31 Biology

For interacting populations,

$$n_{t+1} = F(n_t, u_t, \varepsilon_t), \quad (159)$$

the robust viability kernel describes population states from which management can preserve specified survival conditions.

Common-mode shocks correspond to environmental disturbances, while asymmetric modes correspond to competitive or cooperative divergence among populations.

This analogy reinforces the preventive principle that a system need not have crossed a catastrophic boundary before intervention becomes rational.

32 Psychology and Psychiatry

Latent-state models are ubiquitous in behavioral science.

An abstract model is

$$z_{t+1} = F(z_t, x_t, \eta_t), \quad (160)$$

$$Y_t = H(z_t, x_t) + \nu_t. \quad (161)$$

The present framework emphasizes that a latent representation is not automatically identified merely because it predicts observations well.

The mathematical requirement remains observability or an appropriate statistical identification condition.

Thus

$$\text{prediction} \neq \text{identification}. \quad (162)$$

This distinction is essential whenever latent constructs are interpreted substantively.

33 Political Science and Institutions

Institutional legitimacy can be represented by a constraint

$$L(s, u) \geq 0. \quad (163)$$

Constitutional, legal, accountability, transparency, consent, and procedural constraints can therefore enter the admissible control set.

The optimization problem becomes

$$\mathcal{A}_{\text{inst}} = \{(s, u) : \Phi = 0, C_j \geq 0, L \geq 0\}. \quad (164)$$

This prevents the mathematical optimizer from treating institutionally inadmissible actions as feasible merely because they improve a numerical objective.

34 Geopolitics, Diplomacy, and International Relations

The three-Republic system naturally represents a three-actor strategic network.

A directed cycle

$$A \rightarrow B \rightarrow C \rightarrow A \quad (165)$$

can appear in the strategic Jacobian.

Its feedback strength is represented by products of cross-effects. Such cycles are relevant to alliance formation, arms races, trade disputes, sanctions, technological competition, strategic signaling, and diplomatic escalation.

Diplomatic intervention can therefore be interpreted as a control that modifies the strategic-response Jacobian:

$$\rho(D_u R) \longrightarrow \rho(D_u R + a\text{-effect}). \quad (166)$$

The preventive objective is to move the system away from strategic singularity without unnecessarily disturbing welfare.

35 Warfare Economics and Strategic Survival

In warfare economics, resources may be allocated among consumption, investment, infrastructure, and defense:

$$Y_i = C_i + I_i + G_i + M_i. \quad (167)$$

The control M_i can affect both welfare and strategic survival.

A security-adjusted welfare function can be written abstractly as

$$W_i = U_i(C_i, I_i) + S_i(M_i, s_{-i}) - D_i. \quad (168)$$

The mathematical framework does not prescribe the ethical value of military expenditure. It provides a structure in which the consequences of different allocations can be evaluated subject to explicit welfare and survival constraints.

36 Statistics and Econometrics

The empirical implementation should distinguish parameter uncertainty, latent-state uncertainty, measurement error, stochastic disturbances, and model uncertainty.

The estimated object is therefore not merely

$$\hat{\theta}. \quad (169)$$

It should include

$$\left(\hat{\theta}_t, \hat{z}_t, \hat{\mathcal{D}}_t, \hat{\mathcal{P}}_I^{(H)}\right). \quad (170)$$

Bootstrap, Bayesian, state-space, particle-filter, simulation-based, or other estimators may be employed according to the data-generating process.

No statistical procedure can recover information that is absent from the observation structure.

37 Data Science and Artificial Intelligence

A learned latent-state estimator may have the form

$$\hat{z}_t = f_\varphi(Y_{0:t}), \quad (171)$$

and a learned transition model may be

$$\hat{s}_{t+1} = T_\varphi(\hat{s}_t, a_t). \quad (172)$$

A structural learning objective may combine prediction and certification losses:

$$\mathcal{L} = \mathcal{L}_{\text{prediction}} + \lambda_1 \mathcal{L}_{\text{equilibrium}} + \lambda_2 \mathcal{L}_{\text{stability}} + \lambda_3 \mathcal{L}_{\text{identification}} + \lambda_4 \mathcal{L}_{\text{welfare}}. \quad (173)$$

However, predictive performance does not establish structural validity.

The implementation should therefore treat AI as an estimator, emulator, or policy-search mechanism inside a mathematically constrained architecture.

38 Algorithmic Certification

The complete certification procedure is:

1. Input observations, structural equations, parameter domains, transition maps, strategic maps, and welfare functions.
2. Estimate or characterize latent states.
3. Solve the algebraic equilibrium.
4. Solve the strategic fixed point.
5. Calculate the strategic Jacobian.
6. Construct the strategically reduced state Jacobian.
7. Decompose the system into common and asymmetric modes.
8. Normalize uncertain parameters to $[-1, 1]^d$.
9. Evaluate the model at Chebyshev nodes.
10. Construct Chebyshev expansions.
11. Increase polynomial degree until convergence criteria are met.
12. Construct parameter-dependent Lyapunov matrices.
13. Test positive-definiteness.
14. Test Lyapunov decrease.
15. Compute approximation and numerical error bounds.
16. Calculate strategic, dynamic, welfare, algebraic, and identification margins.
17. Estimate finite-horizon regime-transition probabilities.
18. Test robust viability.
19. If all conditions pass, issue a certificate.
20. If margins are positive but deteriorating, enter preventive-warning mode.
21. Solve the minimum-cost preventive control problem if required.
22. Recompute the entire certificate after intervention.

39 Pseudocode

INPUT:

data Y
state model T
strategic map R
algebraic constraints Φ
welfare functions W_i
parameter domain Θ
shock model P

1. Estimate latent state z .
2. Solve $\Phi(s,u;\theta) = 0$.
3. Solve $u = R(u,s;\theta)$.
4. Compute $K = D_u R$.
5. Compute
$$J = T_s + T_u (I-K)^{-1} R_s.$$
6. Construct common and asymmetric projections.
7. Compute J_C and J_A .
8. Normalize Θ to $[-1,1]^d$.
9. Evaluate J at Chebyshev nodes.
10. Construct Chebyshev coefficients.
11. Increase degree N until convergence.
12. Search for P_C and P_A .
13. Construct Q_C and Q_A .
14. Verify positivity margins.
15. Bound approximation and numerical errors.
16. Compute $m_S, m_C, m_A, m_I, m_{\Delta}, m_W$.
17. Compute transient amplification.
18. Estimate regime-transition probabilities.
19. Test robust viability.

IF all certificate conditions pass:
 return CERTIFIED_SAFE.

ELSE IF all stability margins remain positive:
 return PREVENTIVE_WARNING.

 solve minimum-cost intervention.

 recompute all conditions.

ELSE:
 return INSTABILITY_OR_UNDETERMINED.

OUTPUT:

equilibrium
strategic response
latent state
diagnostic vector
Lyapunov certificates
uncertainty bounds
regime probabilities
viability status
preventive intervention
reproducibility record

40 Unit-Test Architecture

A production implementation should contain at least the following tests.

Test	Required property
Chebyshev recurrence	$T_{n+1} = 2xT_n - T_{n-1}$
Chebyshev boundedness	$ T_n(x) \leq 1$
Node mapping	$x_j \in [-1, 1]$
Coefficient reconstruction	Convergence with increasing N
Clenshaw evaluation	Agreement with direct evaluation
Common projection	$\Pi_C^2 = \Pi_C$
Asymmetric projection	$\Pi_A^2 = \Pi_A$
Orthogonality	$\Pi_C \Pi_A = 0$
Strategic multiplier	Equation (17)
Lyapunov positivity	$P \succ 0$
Lyapunov decrease	$Q \succ 0$
Error correction	Declared bound dominates residual
Warning detection	Warning precedes boundary breach
Control feasibility	Intervention satisfies constraints
Post-control certification	Certificate recomputed successfully
Reproducibility	Same inputs give same certificate

41 Adversarial Testing

The implementation should deliberately test cases in which naive methods fail.

41.1 Near strategic singularity

Choose

$$\rho(D_u R) \lesssim 1. \quad (174)$$

The implementation should report strategic fragility even if the state Jacobian remains stable.

41.2 Near dynamic instability

Choose

$$\rho(J_C) \lesssim 1 \quad (175)$$

or

$$\rho(J_A) \lesssim 1. \quad (176)$$

The corresponding preventive diagnostic must activate.

41.3 Stable but non-normal system

Choose

$$\rho(J) < 1 \quad (177)$$

while

$$\max_h \|J^h\| \gg 1. \quad (178)$$

The implementation must distinguish asymptotic stability from transient resilience.

41.4 Identification failure

Choose

$$\sigma_{\min}(\mathcal{O}_z) \approx 0. \quad (179)$$

The model must not report a highly precise latent-state estimate without qualification.

41.5 Chebyshev under-resolution

Use too small a polynomial degree.

The convergence test must reject the approximation rather than silently certify it.

42 Reproducibility Record

Every certificate should preserve

$$\mathcal{R}_{\text{cert}} = (\Theta, N, M, \{J_{\mathbf{k}}\}, \{P_{\mathbf{k}}\}, \epsilon_J, \epsilon_P, \epsilon_Q, \gamma_C, \gamma_A, \gamma_P, \text{solver information}). \quad (180)$$

This permits independent recomputation.

A production certificate should therefore be treated as a data object rather than merely a textual claim.

43 The Certificate Hierarchy

The framework establishes a hierarchy:

$$\begin{array}{ccccccc} \text{pointwise} & & & & & & \\ \text{numerical} & \prec & \text{grid} & \prec & \text{Chebyshev} & \prec & \text{error-bounded} \\ \text{stability} & & \text{verification} & & \text{approximation} & & \text{Chebyshev} \\ & & & & & & \text{certificate} \end{array} \quad (181)$$

The final level is the strongest because it explicitly controls the difference between the approximation and the true model.

44 The Preventive Principle

The fundamental operational rule is

$$\boxed{\text{intervene before the certificate margin reaches zero.}} \quad (182)$$

More precisely, if

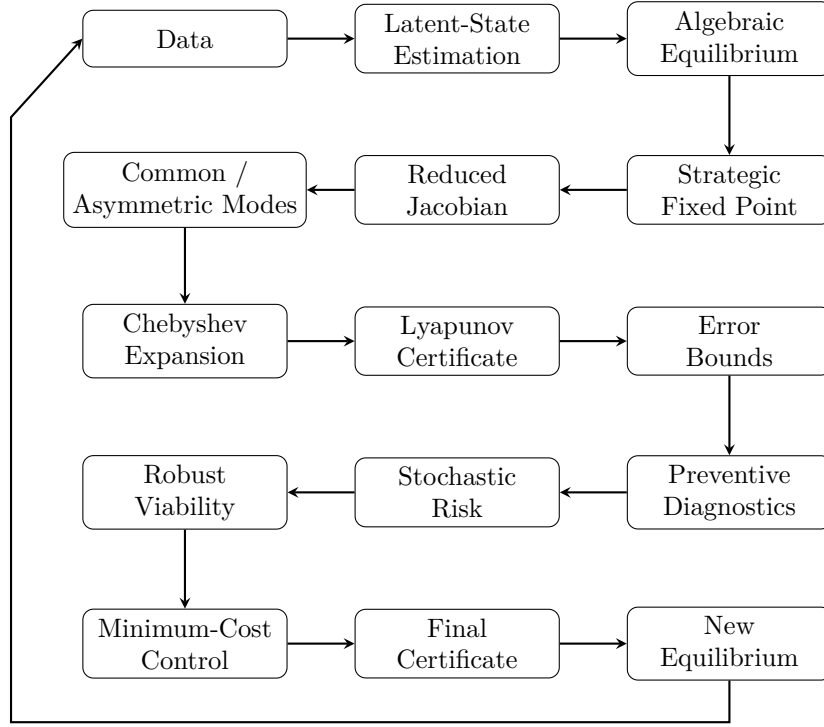
$$\gamma_j(t) > 0 \quad (183)$$

but

$$\dot{\gamma}_j(t) < 0 \quad (184)$$

and the projected crossing lies within the policy horizon, intervention may be justified. The system therefore moves from reactive stabilization toward preventive control.

45 The Complete Computational Architecture



The final feedback loop is therefore

$$\begin{aligned}
 &\text{Data} \rightarrow \text{Identification} \rightarrow \text{Equilibrium} \rightarrow \text{Strategy} \rightarrow \text{Dynamics} \rightarrow \text{Modes} \rightarrow \text{Chebyshev} \rightarrow \text{Lyapunov} \\
 &\rightarrow \begin{matrix} \text{Error} \\ \text{Bounds} \end{matrix} \rightarrow \text{Diagnostics} \rightarrow \text{Risk} \rightarrow \text{Viability} \rightarrow \text{Control} \rightarrow \text{Certificate} \rightarrow \begin{matrix} \text{New} \\ \text{Equilibrium} \end{matrix} \rightarrow \text{Data}.
 \end{aligned} \tag{185}$$

46 The Chebyshev–Lyapunov Theorem

Theorem 46.1 (Chebyshev–Lyapunov Robust Three-Republic Theorem). *Consider the unified three-Republic system with strategically reduced Jacobian $J(x)$ over a compact normalized uncertainty domain $[-1, 1]^d$.*

Suppose there exist finite Chebyshev approximations

$$\hat{J}_N(x), \quad \hat{P}_{C,N}(x), \quad \hat{P}_{A,N}(x), \tag{186}$$

with certified uniform approximation errors.

Suppose further that the common and asymmetric Lyapunov matrices satisfy error-corrected positive-definiteness and Lyapunov-decrease inequalities uniformly over the uncertainty domain.

Suppose also that:

1. *the algebraic equilibrium restrictions are satisfied;*
2. *the strategic equilibrium is uniquely defined;*
3. *the welfare-strategic second-order conditions hold;*
4. *the latent state is uniformly locally identifiable;*
5. *the prescribed diagnostic margins exceed their safety thresholds;*
6. *and the robust stochastic viability condition holds.*

*Then the three-Republic system possesses a machine-checkable robust stability certificate.
If the robust stochastic safety probability satisfies*

$$\Pr[s_t \in \mathcal{S}_{\text{safe}} \forall t \geq 0] \geq 1 - \alpha, \quad (187)$$

then the system possesses a robust preventive safety certificate.

If a feasible minimum-cost intervention restores the required inequalities before any instability boundary is crossed, the system can be returned to the certified safe region without requiring a transition into Regimes II, III, or IV.

Proof. The Chebyshev approximation and its error bounds establish uniform approximation of the relevant matrix functions. The error-corrected positive-definiteness conditions therefore imply positivity of the exact Lyapunov matrices and their decrease matrices. The Lyapunov stability theorem gives uniform exponential stability for the common and asymmetric modes. Strategic contraction establishes a unique strategic response, while the strategic reduction formula incorporates feedback into the dynamic Jacobian. Algebraic feasibility establishes the equilibrium manifold. The welfare conditions establish local welfare-strategic optimality. Uniform observability establishes local latent-state identification. The robust viability condition establishes the existence of an admissible policy preserving the safe set with probability at least $1 - \alpha$. The preventive optimization identifies an intervention whenever its feasible set is nonempty. Therefore all components of the robust certificate hold simultaneously. $\square$

47 Implications for the Unified Three-Republic Programme

The first paper established the equilibrium architecture. The present paper establishes its computational certification architecture.

The distinction can be summarized as

First paper	Present paper
Equilibrium existence	Certificate computation
Strategic stability	Strategic amplification diagnostics
Dynamic stability	Lyapunov certification
Latent-state identification	Identification margins
Robustness	Chebyshev uncertainty representation
Preventive diagnostics	Certified warning thresholds
Preventive control	Certificate-constrained intervention
Theory	Reproducible implementation

The complete theoretical programme is consequently

$$\boxed{\begin{array}{c} \text{Unified} \\ \text{Equilibrium} \end{array} + \begin{array}{c} \text{Chebyshev-Lyapunov} \\ \text{Certification} \end{array} + \begin{array}{c} \text{Preventive} \\ \text{Control} \end{array} = \begin{array}{c} \text{Robust} \\ \text{Three-Republic.} \\ \text{System} \end{array}} \quad (188)$$

48 Conclusion

The central contribution of this paper is the conversion of a continuous robust-stability problem into a finite computational certification problem.

The strategically reduced dynamics are

$$D_s T^* = D_s \tilde{T} + D_u \tilde{T} (I - D_u R)^{-1} D_s R. \quad (189)$$

The parameter dependence is represented by Chebyshev expansions,

$$J(x) \approx \sum_{\mathbf{k}} J_{\mathbf{k}} T_{\mathbf{k}}(x), \quad (190)$$

and parameter-dependent Lyapunov matrices,

$$P(x) \approx \sum_{\mathbf{k}} P_{\mathbf{k}} T_{\mathbf{k}}(x). \quad (191)$$

The stability certificate is

$$P(x) \succ 0, \quad P(x) - J(x)^\top P(x) J(x) \succ 0. \quad (192)$$

The preventive diagnostic system supplements this with

$$m_S, \quad m_C, \quad m_A, \quad \gamma_C, \quad \gamma_A, \quad m_I, \quad m_\Delta, \quad m_W, \quad \mathcal{A}_H. \quad (193)$$

The stochastic layer supplies

$$\mathcal{P}_I^{(H)} = \begin{pmatrix} P_{I \rightarrow II}^{(H)} \\ P_{I \rightarrow III}^{(H)} \\ P_{I \rightarrow IV}^{(H)} \end{pmatrix}. \quad (194)$$

The control layer solves

$$\min_a C(a) \quad (195)$$

subject to restoration of the complete certificate.

The final architecture is therefore not merely a stability test. It is a closed-loop system for

identification $\rightarrow$ equilibrium $\rightarrow$ diagnosis $\rightarrow$ forecast $\rightarrow$ certification $\rightarrow$ prevention $\rightarrow$ re-certification.

(196)

The fundamental methodological principle is:

A system should be certified while it is safe, not merely diagnosed after it becomes unstable.

(197)

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Glossary

Asymmetric mode

A perturbation in the subspace orthogonal to the common Republic vector.

Chebyshev coefficient

A coefficient multiplying a Chebyshev basis polynomial in a finite or infinite expansion.

Chebyshev node

A canonical interpolation or quadrature point associated with the cosine distribution on $[-1, 1]$.

Chebyshev–Lyapunov certificate

A robust stability certificate constructed using a parameter-dependent Lyapunov function represented in a Chebyshev basis.

Common mode

A perturbation shared by the three Republics.

Diagnostic margin

A numerical measure of distance from a specified failure boundary.

Lyapunov function

A positive-definite scalar function that decreases along system trajectories.

Lyapunov matrix

The positive-definite matrix defining a quadratic Lyapunov function.

Observability

The ability to recover internal system states from observable histories.

Preventive control

An intervention undertaken before an instability boundary is crossed.

Robust viability kernel

The set of states from which an admissible policy can preserve safety under the specified uncertainty class.

Strategic feedback multiplier

The matrix $(I - D_u R)^{-1}$ that aggregates repeated strategic reactions.

Transient amplification

Temporary growth in perturbation magnitude despite eventual asymptotic stability.

Uniform certificate

A mathematical certificate that holds throughout the entire prescribed uncertainty domain.

Warning state

A dynamically stable state whose diagnostic margins have deteriorated sufficiently to warrant preventive action.

The End